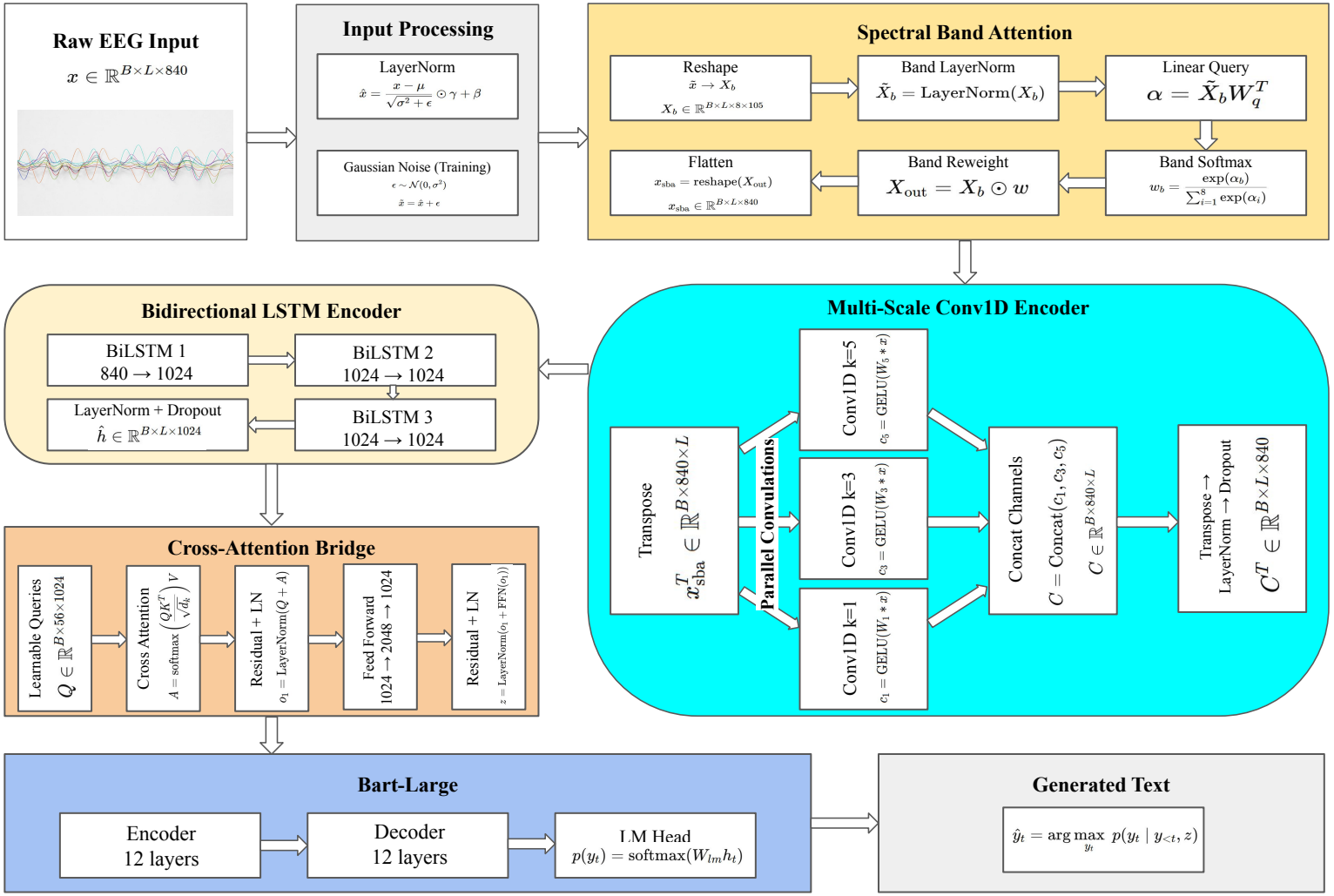
Raw EEG Input
$x \in \mathbb{R}^{B\times L\times 840}$
Input Processing
LayerNorm
$\hat{x} = \frac{x-\mu}{\sqrt{\sigma^2+\epsilon}} \odot \gamma + \beta$
Gaussian Noise (Training)
$\epsilon \sim \mathcal{N}(0, \sigma^2)$
$\tilde{x} = \hat{x} + \epsilon$
Spectral Band Attention
Reshape
$\tilde{x} \to X_b$
$X_b \in \mathbb{R}^{B\times L\times 8\times 105}$
Band LayerNorm
$\tilde{X}_b = \mathrm{LayerNorm}(X_b)$
Linear Query
$\alpha = \tilde{X}_b W_q^T$
Band Softmax
$w_b = \frac{\exp(\alpha_b)}{\sum_{i=1}^{8}\exp(\alpha_i)}$
Band Reweight
$X_{\mathrm{out}} = X_b \odot w$
Flatten
$x_{\mathrm{sba}} = \mathrm{reshape}(X_{\mathrm{out}})$
$x_{\mathrm{sba}} \in \mathbb{R}^{B\times L\times 840}$
Multi-Scale Conv1D Encoder
Transpose
$x_{\mathrm{sba}}^T \in \mathbb{R}^{B\times 840\times L}$
Parallel Convulations
Conv1D k=5
$c_5 = \mathrm{GELU}(W_5 * x)$
Conv1D k=3
$c_3 = \mathrm{GELU}(W_3 * x)$
Conv1D k=1
$c_1 = \mathrm{GELU}(W_1 * x)$
Concat Channels
$C = \mathrm{Concat}(c_1, c_3, c_5)$
$C \in \mathbb{R}^{B\times 840\times L}$
Transpose → LayerNorm → Dropout
$C^T \in \mathbb{R}^{B\times L\times 840}$
Bidirectional LSTM Encoder
BiLSTM 1
840 → 1024
BiLSTM 2
1024 → 1024
BiLSTM 3
1024 → 1024
LayerNorm + Dropout
$\hat{h} \in \mathbb{R}^{B\times L\times 1024}$
Cross-Attention Bridge
Learnable Queries
$Q \in \mathbb{R}^{B\times 56\times 1024}$
Cross Attention
$A = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$
Residual + LN
$o_1 = \mathrm{LayerNorm}(Q + A)$
Feed Forward
1024 → 2048 → 1024
Residual + LN
$z = \mathrm{LayerNorm}(o_1 + \mathrm{FFN}(o_1))$
Bart-Large
Encoder
12 layers
Decoder
12 layers
LM Head
$p(y_t) = \mathrm{softmax}(W_{lm} h_t)$
Generated Text
$\hat{y}_t = \arg\max_{y_t} p(y_t \mid y_{<t}, z)$